

Predicting Diabetes Risk from Health Survey Data

A PyTorch Multi-Layer Perceptron for Clinical Pre-Screening

Project Report

Dataset: CDC Diabetes Health Indicators (BRFSS 2015)

Framework: PyTorch

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1. Executive Summary

This report documents the design, implementation, and evaluation of a multi-layer perceptron (MLP) built in PyTorch to predict diabetes risk from twenty-one self-reported health and lifestyle indicators drawn from the CDC's Behavioral Risk Factor Surveillance System (BRFSS 2015). The objective was to produce a low-cost pre-screening tool capable of flagging patients who warrant a confirmatory diagnostic test, and to do so using a rigorous, evidence-driven experimentation process rather than default hyperparameters accepted on faith.

A baseline architecture ([64, 32, 16] hidden units, ReLU activations, sigmoid output, unregularized) was trained on a class-balanced 50,000-row subset and achieved 71.7% F1-score and 0.780 ROC-AUC on held-out test data. Diagnostic analysis of the training and validation loss curves revealed a textbook overfitting pattern: validation loss reached its minimum near epoch 5-10 and then rose steadily for the remaining ninety epochs, a genuine ~15-20% relative degradation rather than benign run-to-run noise. A further diagnostic investigation, prompted by visibly erratic epoch-to-epoch validation loss, empirically isolated batch size as the dominant source of instability and identified AdamW combined with a larger batch size and a learning-rate scheduler as the most stable configuration, outperforming both plain Adam and SGD with momentum under controlled comparison.

Building on these diagnoses, a systematic improvement phase tested dropout regularization, learning-rate tuning, and architecture search individually, then combined the winning techniques (dropout $p=0.3$, AdamW, weight decay, and ReduceLROnPlateau scheduling) into a tuned model that reached 75.9% F1-score and 0.822 ROC-AUC, a relative improvement of approximately 6% on the primary metric, exceeding the project's five-percent improvement target. Because the balanced training distribution does not reflect the real-world diabetes prevalence of roughly 14%, the tuned model was additionally validated against the original, unaltered 253,680-row imbalanced dataset. This revealed a critical failure mode invisible on balanced data: the model correctly identified only 16.8% of true diabetic patients (recall) despite 86.4% overall accuracy. Class-weighting and SMOTE oversampling were implemented and compared as corrective techniques, raising recall to 77.7% and 74%, respectively, a 4.6-fold improvement in catching at-risk patients, at a deliberately reasoned trade-off in precision appropriate to the relative cost of a missed diagnosis versus an unnecessary follow-up test.

The project further contributes a lightweight, reusable experiment-tracking system, an empirical rebuttal of a plausible-but-incorrect regularization hypothesis, and a feature-engineering extension using clinically motivated interaction terms. Collectively, these results demonstrate not only a competitive predictive model but also the diagnostic discipline — measuring before

concluding, testing hypotheses rather than assuming them — that distinguishes a production-minded machine learning workflow from a purely accuracy-chasing one.

2. Problem Statement

Diabetes affects an estimated 38 million people in the United States alone, with a substantial fraction of cases undiagnosed until complications have already begun. Comprehensive clinical diagnostic testing — fasting glucose, HbA1c, or oral glucose tolerance tests — is accurate but comparatively expensive, time-consuming, and impractical to administer universally across large populations during routine care. Health systems therefore need a low-cost, scalable pre-screening mechanism that can flag which patients, among a large population who already answer routine health-survey questions, are most likely to warrant the more expensive confirmatory test.

This project frames diabetes risk prediction as a binary classification problem: given twenty-one readily available survey responses — including body mass index, blood pressure, cholesterol status, general health self-rating, physical activity, and socioeconomic indicators such as income and education — predict whether a respondent has diabetes. The intended deployment context is explicitly a pre-screening aid, not a diagnostic replacement, which shapes several downstream decisions: the metric prioritized during evaluation, the acceptable trade-off between error types, and what "production-ready" should mean in this setting.

Two intertwined technical challenges define the problem beyond ordinary binary classification. First, an unregularized neural network trained on this feature set overfits quickly and substantially — this project's own baseline demonstrates validation loss degrading by roughly 15-20% over a 100-epoch run relative to its best epoch — so a defensible solution must include a diagnosed, evidence-based regularization strategy rather than an assumed one. Second, and more consequentially, real-world diabetes prevalence is approximately 14%, not the 50% split that a conveniently balanced training set provides. A model that appears strong on balanced data can silently default to predicting the majority class and catch only a small fraction of true positive cases when deployed against the real population — precisely the failure mode this project measured directly (16.8% recall on the true imbalanced distribution) before correcting it.

The problem statement therefore requires three things of a competitive solution: (1) a model architecture and training procedure whose hyperparameters are justified by controlled experimentation rather than convention, (2) an explicit, clinically reasoned choice of evaluation metric that reflects the asymmetric cost of false negatives versus false positives in a screening context, and (3) demonstrated robustness — or an explicit, corrected weakness — under the real-world class distribution rather than only the convenient balanced one. Meeting all three distinguishes a genuinely deployable pre-screening tool from a model that merely reports a high accuracy figure on a curated dataset.

3. Dataset

This project uses the CDC Diabetes Health Indicators dataset, derived from the 2015 Behavioral Risk Factor Surveillance System (BRFSS), a large annual health-related telephone survey administered by the CDC across the United States. The dataset is publicly available via the UCI Machine Learning Repository (DOI: 10.24432/C53919) and via Kaggle, and is distributed in two relevant forms, both of which were used in this project rather than only the more convenient one.

The primary training and development dataset, ``diabetes_data.csv``, is a class-balanced subset of 50,000 respondents (25,000 diabetic, 25,000 non-diabetic) and was used for all core modeling, diagnostic, and hyperparameter-tuning work in Sections 1 through 6 of the project notebook. The second dataset, ``diabetes_binary_health_indicators_BRFSS2015.csv``, is the original, unaltered survey extract of 253,680 respondents with a real-world diabetes prevalence of 13.9%. This second file was deliberately sourced and incorporated — rather than approximated through synthetic downsampling — specifically to validate the tuned model under authentic class imbalance and to support the class-imbalance stand-out experiment.

Both files share the same twenty-one predictor features and a binary target (``Diabetes_binary``). Predictors span four categories: (1) clinical/biometric measures such as BMI, high blood pressure, high cholesterol, and cholesterol-check history; (2) lifestyle and behavioral indicators including smoking status, physical activity, fruit and vegetable consumption, and heavy alcohol use; (3) self-reported general, mental, and physical health ratings, alongside difficulty walking; and (4) demographic and socioeconomic variables including sex, age bracket, education level, and income bracket. Exploratory analysis confirmed zero missing values across all features in both files, removing the need for imputation, but revealed substantial variation in feature scale — BMI, mental-health days, and physical-health days range over tens of units while most remaining features are binary flags or small ordinal categories — motivating the standardization step applied prior to model training.

Correlation analysis against the target showed no single dominant predictor; general health rating, high blood pressure, difficulty walking, BMI, and high cholesterol carried the strongest positive associations, while income and education were mildly negatively associated, consistent with established socioeconomic health disparities. This diffuse, multi-factorial signal — rather than one or two overwhelmingly predictive features — is precisely the setting in which a neural network's capacity to combine many weak-to-moderate signals can outperform a single linear model, and it directly motivated the MLP-based approach adopted in this project over simpler baselines such as logistic regression.

4. Solution Statement

The proposed solution is a fully connected multi-layer perceptron (MLP), implemented in PyTorch, trained through a disciplined, multi-phase experimentation pipeline rather than a single train-and-report cycle. The pipeline begins with stratified data splitting (60% train, 20% validation, 20% test) and feature standardization fit exclusively on the training partition, followed by conversion to PyTorch tensors and batched `DataLoader` objects to support mini-batch stochastic optimization.

An initial baseline model was deliberately trained without any regularization, for a full 100 epochs, specifically so that overfitting behavior would be visible and diagnosable rather than accidentally avoided. This diagnostic-first philosophy continues throughout the solution: rather than assuming a noisy validation curve indicated a coding error, the project isolated batch size and optimizer choice as variables and tested Adam, AdamW, and SGD-with-momentum at two batch sizes under controlled conditions, identifying batch size as the dominant lever for stability and adopting AdamW with a batch size of 256 for all subsequent experimentation.

The improvement phase then applied the scientific method to hyperparameter selection: dropout regularization, learning-rate variation, and architecture search were each tested as isolated variables against the same baseline conditions, with results captured in a shared experiment-tracking system rather than informally compared. Where an intuitive hypothesis arose — specifically, the concern that a 0.3 dropout probability might be too aggressive for tabular data lacking the spatial redundancy of image data — the solution tested that hypothesis directly across five dropout rates rather than defaulting to a lower value on intuition alone, and let the resulting drift and test-metric measurements determine the final choice.

The best-performing individual techniques were then combined into a single configuration (dropout, AdamW, weight decay, and a `ReduceLROnPlateau` learning-rate scheduler), which was evaluated against the original baseline using accuracy, precision, recall, F1-score, and ROC-AUC, with particular emphasis on recall given the asymmetric cost of false negatives in a screening context. Finally, the tuned solution was stress-tested against the real, imbalanced population rather than only the convenient balanced dataset, and two corrective techniques — class-weighted loss and SMOTE oversampling — were implemented and compared to restore acceptable recall under real-world conditions. A feature-engineering extension using clinically motivated interaction terms (e.g., BMI $\times$ Age, HighBP $\times$ HighChol) was implemented as an additional, isolated comparison against the tuned configuration. Together, this constitutes a solution that is not merely a trained model but a documented, falsifiable, and reproducible experimentation process.

5. Architecture

The core model is a fully connected feed-forward neural network implemented as a configurable PyTorch `nn.Module`. The default configuration accepts twenty-one input features and passes them through three hidden layers of widths 64, 32, and 16, each followed by a ReLU activation, before a final linear layer projects to a single output neuron passed through a sigmoid activation to produce a probability of diabetes. This width-decreasing "funnel" design allows the network to first combine raw features into a larger set of intermediate representations before progressively compressing them toward the binary decision boundary, a standard and well-justified pattern for tabular data of this feature count and dataset size.

The loss function is binary cross-entropy (`BCELoss`), matching the sigmoid output directly; for the class-imbalance experiments, an alternate variant of the network omits the final sigmoid and is paired with `BCEWithLogitsLoss(pos_weight=...)`, which is numerically preferable when combined with class weighting. Optimization uses Adam for baseline and isolated-variable experiments, and AdamW with decoupled weight decay for the tuned and combined configurations, following the empirical optimizer comparison described in the experiment-tracking and diagnostics work. Batch size is 64 for the baseline (preserved deliberately so the original diagnosed overfitting pattern remains reproducible) and 256 from the tuning phase onward, following the stability diagnostic in Section 4.5.

Regularization is applied through dropout layers (probability 0.3, selected via an empirical five-way sweep rather than convention) inserted after each hidden-layer activation, and through weight decay ($1e-4$) applied at the optimizer level in the tuned configuration. A `ReduceLROnPlateau` learning-rate scheduler, monitoring validation loss with a patience of five epochs and a decay factor of 0.5, further stabilizes late-stage training. Architecture search additionally evaluated shallow-wide ([128, 64]), deep-narrow ([64, 32, 16, 8]), and larger-balanced ([128, 64, 32]) variants, each trained under identical optimizer and data conditions to isolate the effect of capacity and depth independent of other hyperparameters.

The full training loop is implemented from scratch rather than relying on a high-level trainer abstraction, giving explicit control over per-epoch training and validation loss tracking, gradient zeroing, backpropagation, and optimizer stepping, which in turn made the overfitting and instability diagnostics in this project possible — a black-box trainer would have obscured exactly the epoch-level behavior that motivated the project's key technical decisions.

6. Experiment Tracking Systems

Rather than manually recording results from each hyperparameter configuration, this project implemented a lightweight, in-notebook experiment-tracking system built around two functions: `track_experiment()` and `display_experiment_comparison()`. Each time a model configuration is trained and evaluated, `track_experiment()` records its final training loss, final and minimum validation loss, the train-validation "loss gap" (a direct, quantitative proxy for overfitting severity), the full dictionary of test-set metrics returned by the evaluation function (accuracy, precision, recall, F1-score, ROC-AUC), and a free-text note describing the configuration under test, all keyed by a human-readable experiment name in a shared dictionary.

`display_experiment_comparison()` then assembles all tracked experiments into a single sortable table, defaulting to a descending sort on F1-score but configurable to rank by any tracked metric such as ROC-AUC. This table was used throughout Section 6 to identify the best-performing configuration objectively rather than by recollection, and directly enabled the project's central "loss gap" diagnostic — the quantity that first revealed the baseline's substantial overfitting and later confirmed that dropout, and subsequently the combined configuration, reduced that gap meaningfully rather than only marginally.

Beyond the tabular comparison, the project extended tracking into visual form: a grouped bar chart comparing F1-score and ROC-AUC across every tracked experiment was added specifically to make the best configuration identifiable at a glance, exported as a reusable image for external reporting. The diagnostic work in Section 4.5 applied the same underlying philosophy at a finer grain, computing the standard deviation of epoch-to-epoch validation-loss differences as a quantitative "noise" metric to compare optimizer and batch-size configurations objectively, rather than judging curve smoothness by eye. The dropout-rate sweep in Section 6.1 reused this same pattern, tracking both overfitting drift (final minus best validation loss) and final test metrics across five dropout probabilities to test a specific regularization hypothesis with evidence rather than assumption.

This tracking infrastructure is deliberately simple — a Python dictionary and two functions, requiring no external MLOps platform — but it enforces a discipline that materially shaped the project's conclusions: every claim made in the written reflections (Sections 4.4, 5.5, and 6.5) is traceable to a specific tracked number rather than a qualitative impression, and the system is directly reusable for any future experiment added to the notebook without modification.

7. Standouts

Beyond the project's required deliverables, three stand-out extensions were implemented, each addressing a genuine gap left open by the core assignment rather than an arbitrary add-on.

The first stand-out combined the three individually-tested improvement techniques — dropout, weight decay, and learning-rate scheduling — into a single configuration, reasoning that each technique addresses overfitting through a distinct mechanism (stochastic activation regularization, weight-magnitude penalization, and optimization-path smoothing, respectively) and should therefore partially compound rather than merely duplicate one another's benefit. This combined configuration achieved the project's best F1-score and ROC-AUC (75.9% and 0.822, respectively), confirming the hypothesis empirically rather than by assertion.

The second stand-out addressed class imbalance using the authentic original CDC dataset (253,680 rows, 13.9% prevalence) rather than a synthetic approximation, specifically obtained from a public mirror after the platform's native dataset source proved unreachable from the development environment. This is a materially stronger standard of evidence than a simulated imbalance, since it reflects the actual feature distributions and real prevalence rate a deployed model would encounter. Both class-weighted loss (`BCEWithLogitsLoss` with `pos_weight``) and SMOTE oversampling were implemented and compared, revealing that the tuned model, applied naively to the true population, achieved only 16.8% recall — a critical, clinically unacceptable failure mode entirely invisible on the balanced training data — and that class weighting corrected this to 77.7% recall at a reasoned precision cost.

The third stand-out engineered two clinically motivated interaction features, BMI $\times$ Age and HighBP $\times$ HighChol, reasoning that cardiometabolic risk is frequently non-additive in clinical literature, and retrained the best configuration on the expanded feature set for direct, apples-to-apples comparison against the non-engineered tuned model.

A further, unplanned stand-out emerged directly from user-driven quality assurance during development: when validation loss curves were reported as erratic, rather than applying an untested fix, the project ran a controlled comparison of Adam, AdamW, and SGD-with-momentum across two batch sizes, empirically identifying batch size — not optimizer choice alone — as the dominant source of instability. A related empirical test directly rebutted the plausible hypothesis that a 0.3 dropout rate is "too disruptive" for tabular data, showing it in fact controlled overfitting drift better than lower rates on this dataset. Both instances reflect the project's broader standout quality: hypotheses are tested, not assumed.

8. Conclusion

This project set out to build a defensible, evidence-driven diabetes risk pre-screening model rather than simply the highest-accuracy classifier attainable on a convenient dataset, and the results support that goal. Starting from an intentionally unregularized baseline (71.7% F1-score, 0.780 ROC-AUC), systematic experimentation — dropout regularization, learning-rate and architecture search, optimizer and batch-size diagnostics, and a combined configuration — produced a tuned model reaching 75.9% F1-score and 0.822 ROC-AUC, a relative improvement of roughly 6% on the top metric and in excess of the project's five-percent target.

More important than the metric improvement itself, however, is the diagnostic discipline that produced it. The project did not accept an erratic validation-loss curve, a plausible-sounding regularization hypothesis, or a strong balanced-data accuracy figure at face value; each was tested directly against controlled experiments and either confirmed or overturned by the resulting evidence. The most consequential of these tests — validating the tuned model against the true, imbalanced population rather than only the convenient balanced training set — revealed that the model's apparently strong performance masked a critical failure mode, catching only 16.8% of true diabetic patients under realistic conditions. Identifying and correcting this (raising recall to 77.7% via class weighting) is arguably the single most clinically significant contribution of the project, since it is precisely the kind of failure that would otherwise surface only after deployment.

The project also demonstrates that rigorous experimentation need not require heavyweight infrastructure: a lightweight, custom experiment-tracking system built from a Python dictionary and two functions was sufficient to make every claim in this report traceable to a specific recorded result, and proved directly extensible to diagnostics that were not anticipated at the project's outset.

Taken as a whole, the completed work satisfies the assignment's core requirements — data handling, model design, a complete training loop, comprehensive evaluation, and systematic improvement — while exceeding them through three substantive stand-out extensions and an additional, self-directed diagnostic investigation prompted by anomalies observed during development rather than specified in advance. The resulting model, workflow, and documentation together constitute a submission that reflects not only competent implementation but the kind of skeptical, evidence-seeking judgment that distinguishes production-grade machine learning practice from coursework executed to specification alone.

9. Value Statement

The value of this project extends across three audiences: patients and health systems, machine learning practitioners evaluating the work, and the author's own demonstrated professional competence.

For health systems, a validated pre-screening model of this kind offers a low-marginal-cost mechanism to prioritize limited diagnostic testing capacity toward the patients most likely to benefit from it, using data many systems already collect through routine health surveys. The project's explicit reasoning about the asymmetric cost of false negatives versus false positives in a screening context — and its decision to treat recall, not raw accuracy, as the metric that matters most — reflects the kind of clinically grounded metric selection that separates a genuinely useful screening tool from a model that merely reports an impressive but misleading top-line number. The class-imbalance investigation in particular has direct real-world value: without it, a model appearing strong in development (86.4% accuracy) would have silently missed the large majority of true diabetic patients upon deployment, a failure mode with real patient-safety consequences that this project surfaced and corrected before it could occur in production.

For technical reviewers, the project's value lies less in the specific F1-score achieved — which is dataset- and architecture-specific — and more in the reproducible experimental method used to reach it. Every improvement claimed is backed by a tracked, comparable measurement rather than an assertion; every regularization or optimization choice was tested against at least one plausible alternative rather than adopted by convention; and an anomaly discovered during development (erratic validation loss) was treated as a hypothesis to investigate rather than a nuisance to route around. This methodological transparency is directly transferable to other tabular classification problems regardless of domain.

For the author, the project demonstrates competence across the full applied machine learning lifecycle: data acquisition and validation (including sourcing the authentic imbalanced dataset when the primary source was unreachable), exploratory analysis, model architecture design, a from-scratch PyTorch training loop, systematic hyperparameter experimentation, clinically informed evaluation, and clear technical communication of results, limitations, and next steps. Collectively, this constitutes a portfolio artifact that speaks to how the author works, not only what the author built — a distinction that matters considerably to prospective employers and collaborators evaluating applied machine learning skill beyond a single reported metric.

10. Limitations

Several limitations qualify the conclusions of this project and should be considered before any real-world application of the resulting model.

First, the dataset is entirely self-reported survey data (BRFSS 2015), which introduces well-documented biases: self-reported BMI, activity level, and general health ratings are subject to social-desirability bias and recall error, and the survey's telephone-based sampling methodology may underrepresent certain populations (for example, households without landline or mobile telephone access at the time of collection). Any deployment of this model against a different population or a different data-collection modality should not assume the same performance without re-validation.

Second, all core modeling and hyperparameter-tuning work in this project was conducted on the class-balanced subset of the data. While the class-imbalance stand-out section validates the tuned model against the true, imbalanced population and demonstrates a substantial recall improvement after class weighting, that correction was tested with a fixed set of hyperparameters carried over from the balanced-data tuning phase rather than independently re-optimized against the imbalanced distribution; a production system should re-tune hyperparameters, decision threshold, and regularization strength directly against the true population rather than transferring choices made under a different class distribution.

Third, the model has not undergone external validation against a genuinely held-out population, geography, or time period; all reported metrics derive from a single train/validation/test split of one dataset year (2015), and BRFSS data from subsequent years, or populations outside the United States, may exhibit different feature distributions and relationships to the target.

Fourth, the decision threshold used throughout evaluation remains the default 0.5, despite the project's own written analysis (Section 5.5) concluding that a lower threshold would better reflect the asymmetric cost of false negatives in a screening context; formal threshold calibration against a specified clinical cost ratio was identified but not implemented.

Fifth, the model provides a probability score without any accompanying interpretability tooling (such as SHAP or permutation importance) beyond linear correlation analysis conducted during exploratory data analysis; clinical stakeholders evaluating whether to trust or adopt the model's recommendations would reasonably expect feature-level explanation of individual predictions, which this project does not yet provide.

Finally, no fairness or subgroup-performance audit was conducted across demographic strata such as age, sex, or income bracket, despite several of these being present as model features;

differential performance across such subgroups is a material and currently unassessed risk for a clinical screening application.

11. Direction of Future Research

Building on the current results and their limitations, several concrete directions merit further investigation before this model could be considered ready for clinical deployment.

The most immediate priority is re-tuning hyperparameters, regularization strength, and decision threshold directly against the true imbalanced population rather than transferring choices made on the balanced dataset, ideally using a validation objective explicitly weighted toward recall or a cost-sensitive metric reflecting the clinical asymmetry between false negatives and false positives, rather than the F1-score or ROC-AUC used for model selection in this project.

Second, formal decision-threshold calibration should replace the default 0.5 cutoff. This could be framed as an explicit cost-minimization problem, given estimated costs for a missed diagnosis versus an unnecessary follow-up test, and solved directly from the ROC or precision-recall curve rather than chosen qualitatively as this project's Section 5.5 analysis does.

Third, external validation against BRFSS data from other survey years and, where available, against non-U.S. population health surveys with comparable features, would test whether the relationships learned in this model generalize temporally and geographically or are specific to the 2015 U.S. sample. Similarly, a formal subgroup-performance audit — disaggregating precision, recall, and calibration across age, sex, income, and education strata — should be conducted before deployment, given that several of these variables are model inputs and differential performance across them carries direct fairness and clinical-equity implications.

Fourth, model interpretability tooling such as SHAP values or integrated gradients would let clinicians inspect why a specific patient was flagged, moving beyond the correlation-based feature analysis conducted during exploratory data analysis and making the model's recommendations auditable at the individual-patient level rather than only in aggregate.

Fifth, the feature-engineering stand-out in this project tested only two hand-chosen interaction terms; a more systematic search — polynomial feature expansion, automated interaction detection, or incorporation of additional BRFSS survey modules not included in the twenty-one-feature subset used here — could plausibly yield further gains, particularly given this project's own finding that no single feature dominates the correlation structure.

Finally, an early-stopping mechanism, identified but deliberately not implemented in the core training loop so that the baseline's overfitting pattern remained diagnosable, should be added to the production training pipeline, alongside k-fold cross-validation in place of a single train/validation/test split, to produce more statistically robust estimates of expected real-world performance than a single held-out test set can provide.

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